A Experimental Details

A.1 Prompt

The prompt template used to generate the data is given in Figure 6.

A.2 Metrics

Sample-Level Metrics To investigate how correctness varies with length in sample level, we sort the N samples for each question in ascending order of reasoning length $l_i^{(q)}$, creating a length-based ranking. For each rank position $r \in \{1, \cdots, N-1\}$, define the mean reasoning length and mean accuracy at rank r across the dataset:

$$L_r = \frac{1}{|D|} \sum_{q \in D} l_r^{(q)}, \ \text{Acc}_r = \frac{1}{|D|} \sum_{q \in D} c_r^{(q)}$$

These metrics allow us to examine whether, across different reasoning paths for the same question, shorter or longer responses tend to yield higher accuracy.

Prompt Template

```
<|begin_of_sentence|><|User|>
    Output the final answer within \boxed{}. [question]
<|Assistant|><think>
```

Figure 6: The prompt template used to generate responses

Question-Level Metrics At the question level, we study the relationship between a question's overall average reasoning length and its overall correctness averaged over N samples. Specifically, for each question q, we compute:

$$L^{(q)} = \frac{1}{N} \sum_{i \in [N]} l_i^{(q)}, \text{Acc}^{(q)} = \frac{1}{N} \sum_{i \in [N]} c_i^{(q)}$$

When accessing how overall response length relates to correctness (as Table 1), we use the average token lengths of correct and incorrect responses, denoted as $L^{\checkmark}$ and and $L^{\times}$, respectively:

$$L^{\checkmark} = \frac{1}{|D^{\checkmark}|} \sum_{q \in D^{\checkmark}} l^{(q)}, \ L^{\times} = \frac{1}{|D^{\times}|} \sum_{q \in D^{\times}} l^{(q)},$$

where $l^{(q)}$ is the response length for question q, and $D^{\checkmark}$ and $D^{\times}$ are the sets of questions answered correctly and incorrectly, respectively.

B Additional Experiments for Sample Level Analysis

B.1 Distribution of Extremes of Rank Indices.

To analyze the distribution of correctness at the extremes of the ranked list for each question, we define the following rank-based indicators:

- $r_{\text{min}}^{\checkmark}(q)$: rank of the shortest correct sample, i.e., $\min\{r | c_r^{(q)} = 1\}$.
- $r_{\text{max}}^{\checkmark}(q)$: rank of the longest correct sample, i.e., $\max\{r | c_r^{(q)} = 1\}$.
- $r_{\text{min}}^{\times}(q)$: rank of the shortest incorrect sample, i.e., $\min\{r | c_r^{(q)} = 0\}$.

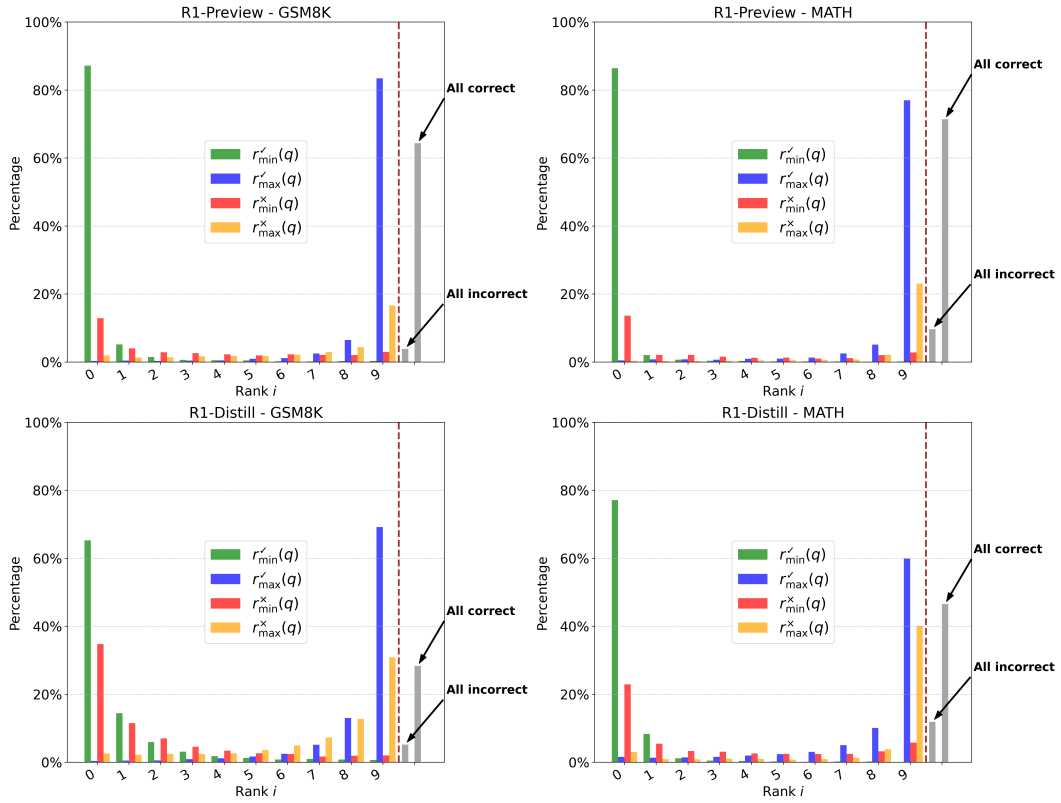

Figure 7: Distribution of the extreme rank indices $r^{\checkmark}_{\min}, r^{\checkmark}_{\max}, r^{\times}_{\min}, r^{\times}_{\max}$. These represent the positions of the shortest and longest correct and incorrect samples in the length-ranked list for each question. The gray bar on the right indicates the proportion of questions for which all samples are either correct or incorrect—cases where the corresponding extreme rank indices do not exist.

- $r^{\times}_{\max}(q)$: rank of the longest incorrect sample, , i.e., $\max\{r | c_r^{(q)} = 0\}$.

$r^{\checkmark}_{\min}(q) = i$ indicates that the i-th shortest sample already contains sufficient reasoning to solve the problem correctly. In contrast, $r^{\checkmark}_{\max}(q) = i$ identifies the longest reasoning length for which correctness is still preserved. When $r^{\checkmark}_{\max}(q) < N-1$, it suggests that any further increase in reasoning length—beyond the i-th length ranked sample—introduces noise or errors that begin to degrade performance. In this sense, $r^{\checkmark}_{\max}(q)$ helps locate the boundary at which reasoning becomes too excessive and starts to erode correctness. Similarly, the distributions of $r^{\times}_{\min}(q) = i$ and $r^{\times}_{\max}(q) = i$ characterize the boundaries of incorrect responses. These distribution of these extremes of rank indices are given in Figure 7.

Correct Answers Often Suffice Within the Shortest Few Samples As shown in Figure 7, over 80% of questions on both GSM8K and MATH when using R1-Preview, the shortest sample is already correct, i.e., $r^{\checkmark}_{\min}(q) = 0$. For R1-Distill, this proportion is slightly lower—over 60% for GSM8K and over 70% for MATH. Focusing on the distribution of $r^{\checkmark}_{\min}(q)$ among questions where $r^{\checkmark}_{\min}(q)$ exist, (i.e., at least one sample is correct), we observe that for the R1-Preview-GSM8K setting, most questions are answered correctly by either the shortest or second shortest sample, with $r^{\checkmark}_{\min}(q) = 0$ and 1 together covering a large majority. For R1-Preview-MATH, $r^{\checkmark}_{\min}(q) = 0$ alone accounts for the majority of answerable questions. In contrast, for R1-Distill-GSM8K, $r^{\checkmark}_{\min}(q) = 0$ represents a smaller proportion compared to the other three settings, and a non-negligible fraction of questions require slightly longer reasoning, with $r^{\checkmark}_{\min}(q) = 1$ and 2 also contributing substantially. The cumulative distribution of $r^{\checkmark}_{\min} \leq i$ in Figure 8 shows the above observation more clearly. Specifically, the plot for $r^{\checkmark}_{\min} \leq i$ in Figure 8 shows the proportion of questions that are answered correctly by the i-th shortest or shorter response. For all settings except R1-Distill–GSM8K, the

cumulative percentage increases rapidly and begins to plateau after $i > 1$, indicating that the top two shortest generations are sufficient to answer nearly all questions the model is capable of solving.

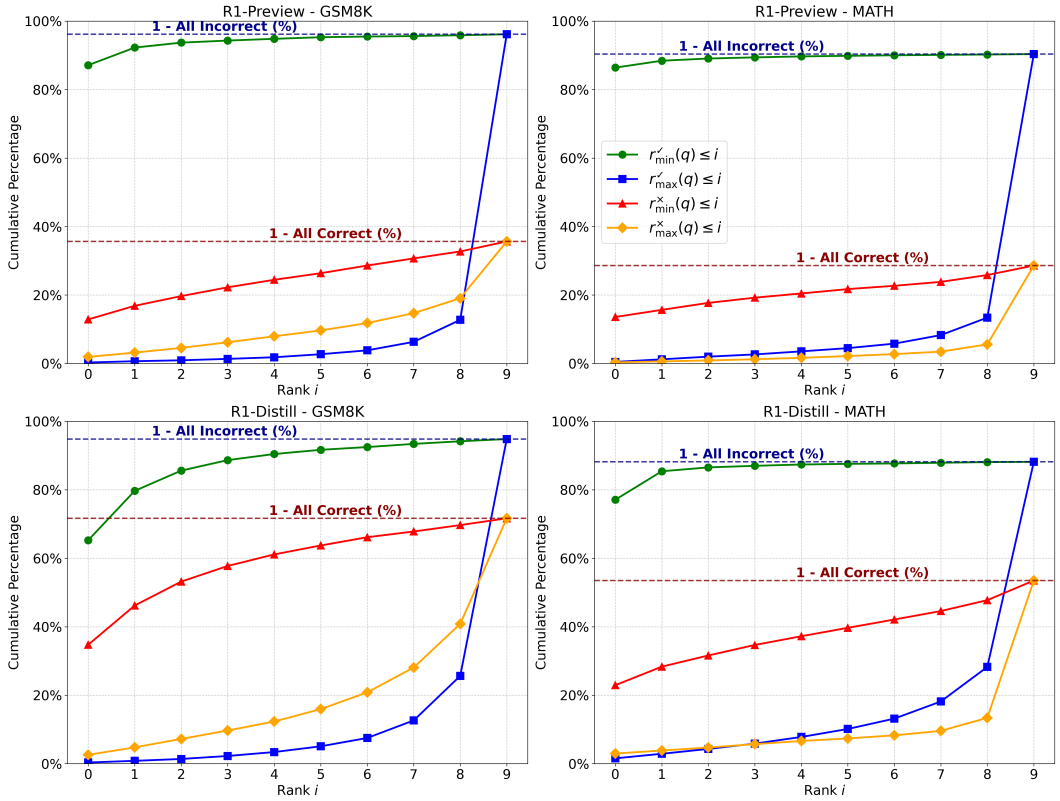

Figure 8: Cumulative distribution the extreme rank indices $r^{\checkmark}_{\min}, r^{\checkmark}_{\max}, r^{\times}_{\min}, r^{\times}_{\max}$. The dashed line labeled "1 - All Incorrect" indicates the proportion of questions with at least one correct sample, while the line labeled "1 - All Correct" marks the proportion of questions where at least one sample is incorrect.

Excessive Reasoning Can Degrade Performance The distribution of $r^{\checkmark}_{\max}(q)$ highlights the point at which increasing reasoning length begins to undermine correctness. Specifically, when $r^{\checkmark}_{\max}(q) = i$ for any $i < N - 1$, it indicates that correctness is maintained up to the i-th ranked sample but is lost at rank $i{+}1$. This suggests that beyond a certain length, additional reasoning introduces errors that compromise an otherwise correct answer. This effect is most apparent near the end of the ranked list—that is, among the longest few samples. For R1-Preview, the proportion of questions with $r^{\checkmark}_{\max}(q) = i$ remains low for $i < N{-}2$, but becomes noticeable at $i = N{-}2$: approximately 5% of questions that are answered correctly at this rank become incorrect at the longer response. For R1-Distill, the issue is more pronounced—around 10% of questions that are correct at rank $N - 2$ become incorrect at rank $N - 1$. These patterns suggest that excessive reasoning can degrade performance by either exceeding the maximum token length or introducing errors that turn an initially correct answer into an incorrect one.

B.2 Correlation of Token Length of Each Rank and the Correctness

In Figure 9, we show the spearman correlation of token length and correctness for responses at each rank, the correlations evaluated on all the questions are on the left subgraph while on the right of the subfigure, we evaluated all the medium questions, i.e., questions that are neither all correct nor all incorrect across all the N samples. We find that, there is a strong correlation with correctness and the length among different samples for the same questions. This indicates that, fixing a question difficulty and the model, where the optimal token required to answer the question would be decided, generating